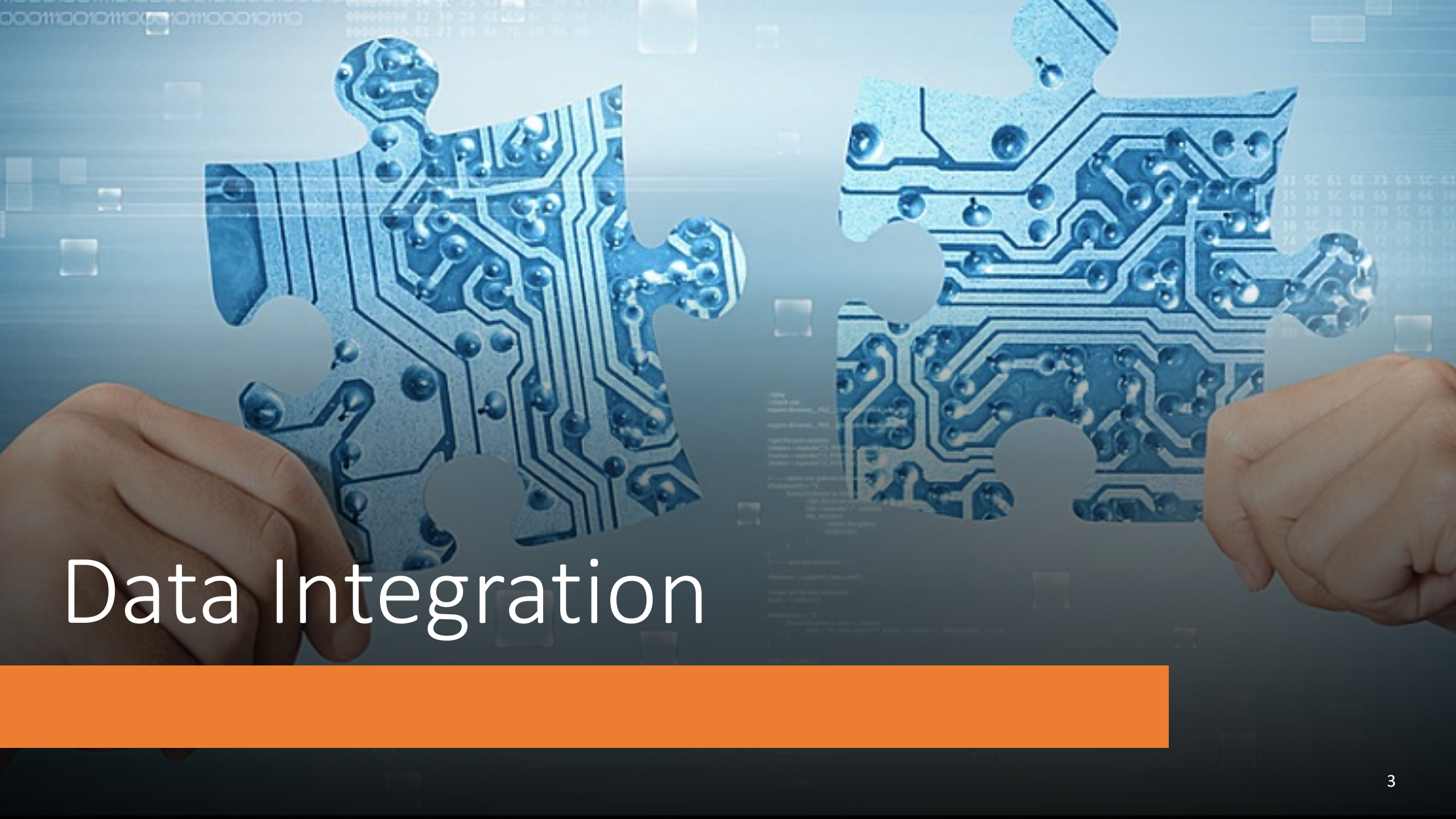


BSc in IT: Specialising in Data Science

IT3021: Data Warehousing
and Business Intelligence

Lecture 05

Data Ingestion Flows



Data Integration

What is Data Integration?

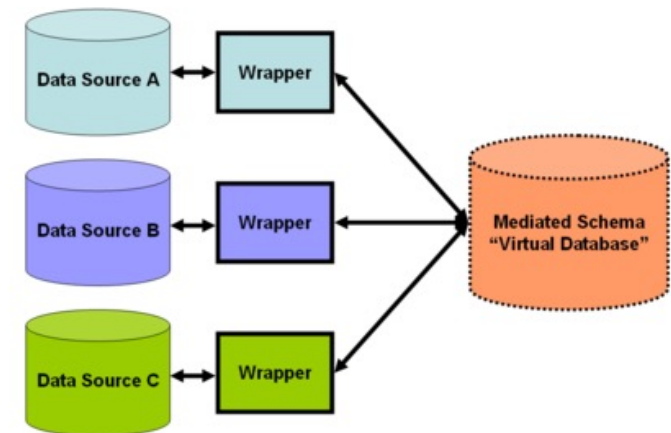
- Data integration is consolidation of business information or data sets from various sources, maybe in various formats, and providing users with a unified view.

Data Integration Techniques

Combines data virtually from multiple sources in real-time without physically moving or copying the data.

- Data Federation

- Provides a single virtual view of one or more data source
- No physical data movement
- Pulls data from source systems on an on-demand basis
 - A query against the virtual view, retrieves data from source systems, integrates the result and sends to the requesting application/user
 - Hence, real time!
 - Required data transformations are done dynamically
 - These details abstracted from the application/user
- Application/user sees only the virtual view (mediated schema)
- Implementation approach:
 - Enterprise Information Integration (EII)



Data Integration Techniques

Pushes data from source to target systems (event-driven), ensuring delivery either synchronously or asynchronously.

- Data Propagation

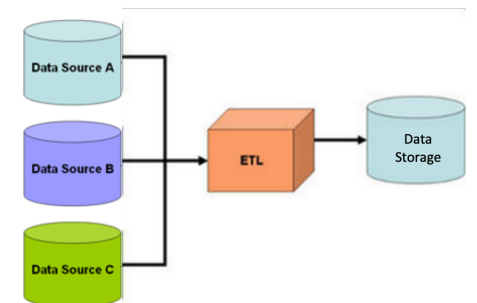
- Usually operate online and push data to the target location
 - Event-driven
- Updates in a source system may be propagated asynchronously or synchronously to the target system
 - Synchronous propagation requires that updates to both source and target systems occur in the same physical transaction
- Regardless of the type of synchronization, data propagation guarantees the delivery of the data to the target
 - This guarantee is a key distinguishing feature of data propagation
- Implementation approaches:
 - Enterprise Application Integration (EAI): applications copy data from one location to another
 - Enterprise Data Replication (EDR): different mechanisms available

Data Integration Techniques

Combines data from multiple sources into a single storage (like DW, ODS) using ETL or ELT, often with some processing delay.

- Data Consolidation

- Data from multiple source systems are integrated into a single persistent data store
 - Used for reporting, analysis
 - Act as a data source for downstream applications
 - e.g., Data Warehouse, Operational Data Store, Data Mart, Data Lake
- Usually there is a delay/latency for the updates in sources to appear in the target store
 - Latency depends on business needs; seconds, hours, days, etc.
 - Data with zero latency is known as real-time data
 - Difficult to achieve using data consolidation (big data tools & technologies used)
- Implementation approach:
 - Extract-Transform-Load (ETL)
 - Extract-Load-Transform (ELT)



ETL Process

Source Systems



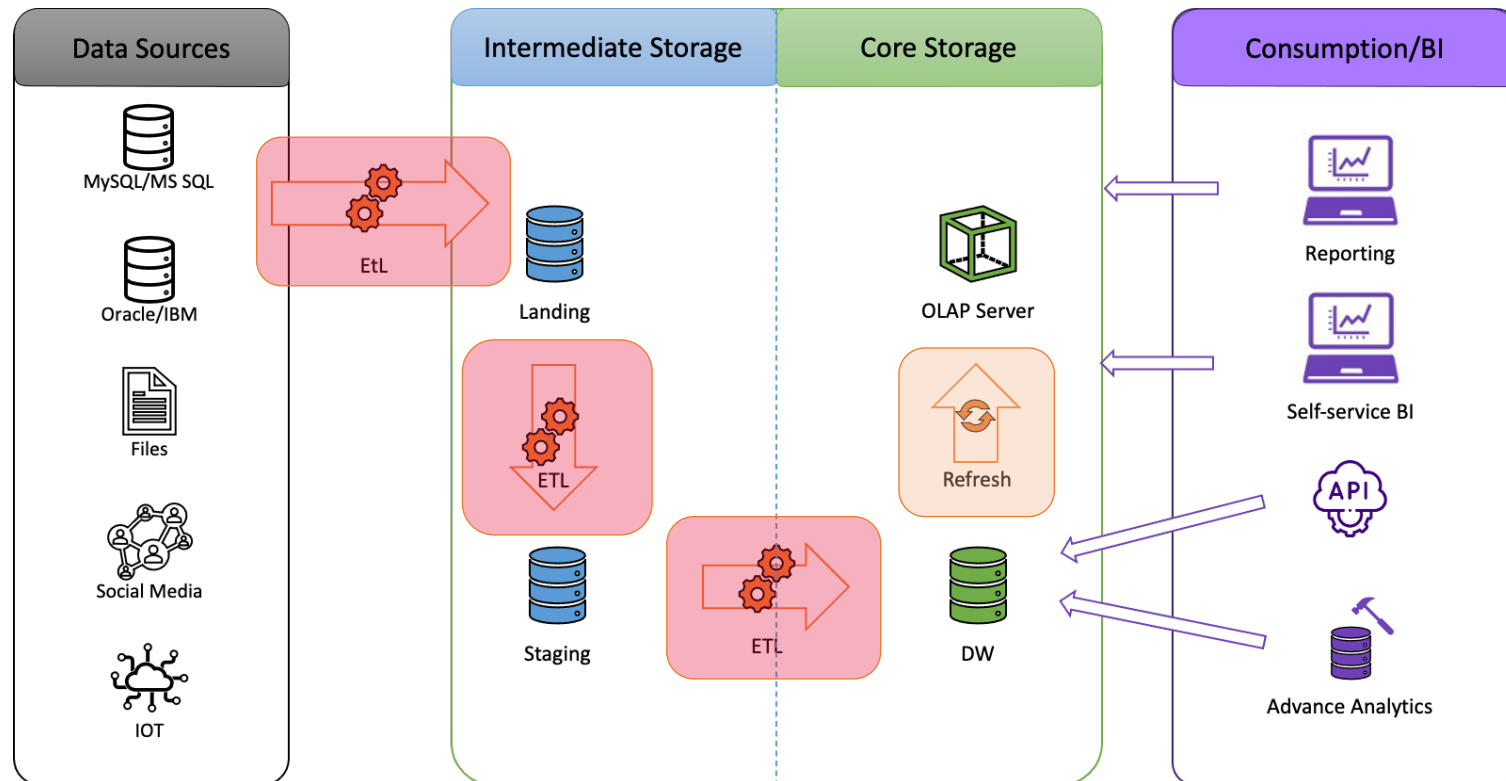
Destination
(Data Warehouse)



Extract, Transform, Load

ETL at Several Levels

- Many ETL layers due to intermediate storage layers
 - Intermediate storage layers are used to minimize dependency and minimize impact on source systems



ETL Considerations

- Different types of sources:
 - Database systems
 - Files (.csv, .txt, .xlsx, etc.)
 - Documents (JSON, XML, etc.)
 - Legacy systems
 - APIs
- Data source connectivity and security concerns
- Different frequency requirements:
 - Near real-time, hourly, daily, weekly, etc.

ETL Considerations

- Extraction window limitations:
 - Small time frames to cater near real time requirements
 - Probably off-peak times for scheduled batch extractions
- Source data realities
 - Data quality issues
 - Extraction window limitations
- Capabilities of tools used
 - Support of traditional ETL tools for real-time extractions

First, Understanding Data!!!

- Before data extraction, need to understand source data
- **Data Profiling** to understand data:
 - Profiling is the technical analysis of data: describe its content, consistency, and structure
 - Strategic decisions: whether to include certain data or not in the data warehouse
 - Tactical decisions: find data issues as much as possible, and try to resolve
 - Issues found will be:
 - Notify/sent back to the originator of the source for correction
 - Resolve as a part of the data quality process

Data Extraction

Logical Extraction Methods

- **Full extraction**

- Extract all the data, every time!
- Some dimensions can be flushed and loaded based on the complexity and requirements

- **Incremental extraction**

- Extract only the changes/updates
- Most dimensions will be loaded with latest changes allowing the maintenance of historical information
- Fact table data

Physical Extraction Methods

- **Online extraction**

- Data extracted directly from the source systems itself

- **Offline extraction**

- Data staged outside the original source system
 - Source system push the data into a sperate storage in some format
 - Flat files
 - Dump files
 - Extraction process access this storage

Push versus Pull

- **Push**

- A process in the source system creates a snapshot of changes delivers to downstream systems
- Mostly occurs in incremental extraction and offline extraction

- **Pull**

- Downstream system connects to the source, request to get data from the source
- Mostly occurs in online extraction
- Could be used with both full and incremental extraction

Incremental Extraction: Capturing Changes

- **Change Data Capture (CDC)**: the process of capturing changes occurred at the source systems in order applying them throughout the enterprise
- CDC minimizes the resources required for ETL processes
 - Because it only deals with data changes

Source table

Client_ID	FName	LName	DoB	MStatus
135001	John	Doe	1/12/1960	married
135002	John	Doe Junior	12/31/1985	single
135003	John Q.	Public	5/6/1985	divorced
135004	Judy	Doe	10/22/1980	single
135005	Jane	Roe	8/16/1982	married

Target table

ClientID	FirstName	LastName	DoB	MStatus	DateBegin	DateEnd	St_Code
135001	John	Doe	1/12/1960	married	2/1/2011		I
135002	John	Doe Junior	12/31/1985	single	2/1/2011		I
135003	John Q.	Public	5/6/1985	divorced	2/1/2011		I
135004	Judy	Doe	10/22/1980	single	2/1/2011		I
135025	Jane	Roe	8/16/1982	married	2/1/2011		I

Figure 1. Initial Load

Source table

Client_ID	FName	LName	DoB	MStatus
135001	John	Doe	1/12/1960	married
135003	John Q.	Public	5/6/1985	divorced
135004	Judy	Smith	10/22/1980	married
135005	Jane	Roe	8/16/1982	married
135006	Joe	Bloggs	7/14/1970	single

Target table

ClientID	FirstName	LastName	DoB	MStatus	DateBegin	DateEnd	St_Code
135001	John	Doe	1/12/1960	married	2/1/2011		I
135002	John	Doe Junior	12/31/1985	single	2/1/2011	2/2/2011	D
135003	John Q.	Public	5/6/1985	divorced	2/1/2011		I
135004	Judy	Doe	10/22/1980	single	2/1/2011	2/2/2011	I
135025	Jane	Roe	8/16/1982	married	2/1/2011		I
135004	Judy	Smith	10/22/1980	married	2/2/2011		U
135006	Joe	Bloggs	25763	single	2/2/2011		I

Figure 2. Incremental Load

Capturing Changes with CDC

- Goals of CDC:
 - Capture all changes made to source data
 - Isolate the changes to allow selective processing
 - Tag changes with reason codes to distinguish error corrections from true updates
 - Support compliance tracking
- **Note:** CDC is not required for the initial data load

Change Data Capture: Approaches

- **Full-diff compare**

- Uses a snapshot of yesterday's data and compares with today's snapshot to find changes on a record by record and field by field basis
- A thorough approach but resource intensive

- **CRC-code** (Cyclic Redundancy Checksum)

- Identifies the changes using the CRC code
- Recognizes the changes more efficient than the field by field compare method

Change Data Capture: Approaches

- **Database log scrapping**

- Takes a snapshot of the database log and extracts modified records

- **Delta file**

- Contains all the data modifications that were made by applications during business operational activities
- Source systems should generate this files
- CDC process become very easy and efficient
- A dependency on the source system!

Change Data Capture: Approaches

- **Trigger method**

- Adding triggers to source tables whose changes should be controlled
- Triggers are turned on for insert, update or delete operations and record these modifications in a specific tables or files; then read the modification from these tables or files

- **Message queue monitoring**

- Used in a message-based operational/transactional systems
- Low overhead
- Monitors the message queue for all the operations/transactions

Change Data Capture: Approaches

- **Timed extracts**

- Typically selects all rows where the source record *create* or *modified* date-time values are greater than or equal to last extracted date-time
 - *SYSDATE – 1* for a daily ETL; but need to ensure failed jobs are retriggered
- Manual intervention is required if the process fails to perform clean-ups
- Tend to have duplicates if the process fails
 - Must be addressed before loading them into the target

Transformations

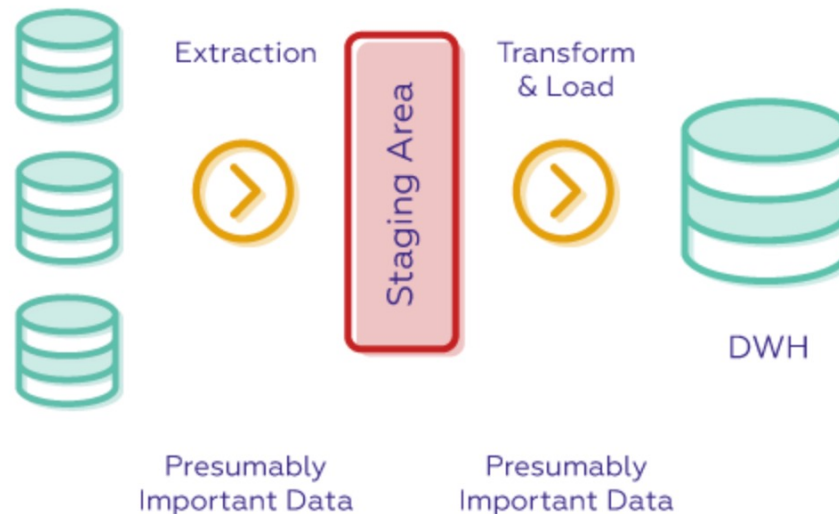
Purpose

- Transform data stored in different data structures to a uniform data structure
- Apply data cleansing approaches to deal with data issues
- Apply business logic to generate expected data set that needs to be stored in data warehouse
- Add value to data by enhancing raw data
- Can be architected to generate metadata that can be used to diagnose what's wrong with source systems
 - Thus, provide suggestions to source systems to correct erroneous data within the source

Transformation Techniques

- **Multistage data transformation (ETL)**

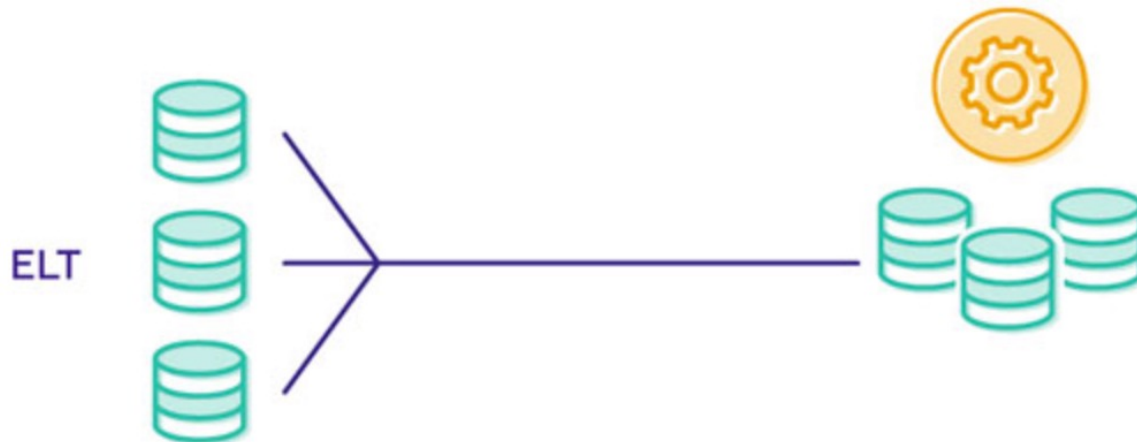
- This is the classic extract, transform, load process
- Extracted data is moved to a staging area where transformations occur prior to loading the data into the warehouse



Transformation Techniques

- **In-warehouse data transformation (ELT)**

- Data is extracted and loaded into the analytics warehouse, and transformations are done there
- Utilize the increased performance and scalability of the modern analytics databases
- ELT tools provide ability to write in-database transformations to be written in SQL



ETL vs. ELT

Parameter	ETL	ELT
Process	Transformation of data happens at the staging server and then loaded to the data warehouse	Data is loaded into the data warehouse and then transformation takes place
Data size vs. complexity of transformations	ETL is best suited for dealing with smaller data sets that require complex transformations	ELT is best when dealing with massive amounts of structured and unstructured data
Added calculations	Calculations will either replace existing columns, or append the dataset to push them to the target data system	ELT adds calculated columns directly to the existing dataset
Implementation complexity	Highly evolved ETL tools are available and experts are easy to find	Tools are still evolving and experts with the right knowledge and skills can be difficult to find

ETL vs. ELT

Parameter	ETL	ELT
Waiting time to load information	ETL load times are longer than ELT because it's a multi-stage process: 1. data loads into the staging area 2. transformations take place 3. data loads into the data warehouse Once the data is loaded, analysis of the information is faster than ELT	Data loading happens faster because there's no waiting for transformations and the data only loads one time into the target data system. However, analysis of the information can be slower than ETL if not pre-prepared
Availability of data in the system	Transforms and loads the data that you decide as necessary during the design. Thus, only this information will be available	ELT can load all data immediately, and users can determine later which data to transform and analyse
Adoption of the technology and availability of tools and experts	Well-developed process used for over 20 years and experts are readily available	As this is a new technology it can be difficult to locate experts and more challenging to develop

ETL vs. ELT

Parameter	ETL	ELT
Costs	On-prem tools are mostly commercial tools and most of them are expensive. Cloud based tools are available with pay-per-usage models	Open source tools are also available, Can load and save data without transforming, then apply transformations as needed which will save money if pay-per-usage cloud services are used
Hardware requirements	Legacy, onsite ETL processes have extensive and costly hardware requirements, but they are not as popular today	ELT processes don't require special hardware and there are many cloud-based solutions too
Compliance	ETL can remove sensitive information before putting it into the data warehouse. Hence, it is easier to satisfy compliance standards. It also protects data from hacks and inadvertent exposure	ELT requires data to be uploaded to the target before removing sensitive information. This could violate compliance standards. Sensitive information will be more vulnerable to hacks and inadvertent exposure

ETL vs. ELT

Parameter	ETL	ELT
Unstructured data support	ETL can be used to structure unstructured data, but it can't be used to pass unstructured data into the target system	ELT is a solution for uploading any type of data (unstructured, semi-structured and structured) into a data lake and make them data available to business intelligence systems
Compatibility with data lakes	Generally transforms data for integration with a structured relational data warehouse system	ELT offers a pipeline for data lakes to ingest unstructured data and then transforms the data on an as-needed basis for analysis

Transformations

- Application of business logic
 - Calculation
 - Derived columns (e.g., *sale_amount = qty x unit_price*)
 - Joining and splitting (e.g., linking data from multiple sources, splitting comma separated values)
- Data cleansing
 - Deduplication
 - Handling NULL values
 - Format unit of measure
 - Format texts
 - Change text to lower, upper or proper case
 - Get rid of extra spaces
 - Data conversion such as converting numbers stored as text into numbers
 - Spell check

Transformations

- Data filtering
 - Selection of required columns
 - Filter unwanted outliers
 - Filter records based on quality or requirement
- Data enrichment
 - Value added data
 - Full name to first, middle and last name
 - Adding social data with available data
 - Adding up all purchases a customer has made to build a customer lifetime value (CLV) metric
- Data validation
 - Simple or complex data validation
 - For example, if the first three columns in a row are empty then reject the row from processing

Transformations

- Generating surrogate-key values
- Look-ups
- Transposing or pivoting
- Aggregation
 - Data elements are aggregated from multiple data sources and databases
 - for example, rollup; summarizing multiple rows of data — total sales for each store, and for each region, etc.

Data Quality

- Handling missing data:
 - Dropping records that have missing values is one approach
 - However, this is sub-optimal because when you drop records with missing data, you drop potentially valuable information
 - The fact that the values were missing may be informative in itself
 - In the real world, you often need to make predictions on new data even if some of the features are missing!
 - Generating missing values based on other observations whenever possible is another approach

Data Quality

- Handling missing **categorical data**
 - Generate missing values using a business logic if possible
 - Else, the best way to handle missing data for categorical features is to simply label them as '*missing*'!
 - This is essentially adding a new class (category) for the feature (attribute)
 - This tells the algorithm that the value was missing
 - This also gets around the technical requirement to handle missing values

Data Quality

- Handling missing **numeric data**
 - Generate missing values using a business logic if possible
 - Else, for missing numeric data, flag and fill the values
 - Flag the records with an indicator
 - Then, fill the original missing value with a suitable numeric value (which is not used as actual data) just to meet the technical requirement to handle missing values
 - By using this technique of flagging and filling, you are essentially allowing algorithms to estimate the optimal constant for missingness (e.g., in ML), instead of just filling it in with the mean value based on available data

Data Quality Systems

- There are dedicated data quality tools to cleanse data
 - Goal is to offer a comprehensive architecture for cleansing data, capturing quality events as well as ultimately controlling data quality
 - Not only limited for data warehousing solutions
- Should start by profiling data to understand the depth, we should address
- These systems can provide specific descriptions of data errors, expected to be found in ETL processes
- These systems can be integrated with ETL process to cleanse data are a separate task

Data Quality Validation

- Quality validation tasks ensure data provided through ETL/ELT processes are in acceptable level
 - Can be considered as a testing task within ETL process
- Two main tasks
 - **Quality screens**
 - Column screen: checks data within a single column (e.g., data type)
 - Structure screen: test relationships of data across columns (e.g., constraints such as keys)
 - Business rule screen: checks data by relating them to business rules
 - **Respond to quality events**
 - Halting the process
 - Sending the error records to a suspense file for later processing
 - Tagging the record with an error code and passing it to the next step

Data Loading

Data Loading Concerns

- Order of data loading
 - Dimensions vs. facts
- Surrogate key generation
- Hierarchy management
- Loading data into special dimensions
 - Slowly changing dimensions
- Fact table loading
 - Transaction vs. periodic snapshot vs. accumulating fact tables
- Loading aggregate tables

Loading into Dimension Tables

- Dimensions provide different aspects to analyse measures in fact tables
- They are usually smaller in size compared to fact tables
- Dimensions are the first to be loaded with latest data before updating the fact tables
 - Fact tables contains references to dimension tables using surrogate keys
- Dimension tables could be loaded in different ways
 - Fresh load
 - delta load
 - Follow different SCD approaches

Loading Slowly Changing Dimensions

- Based on the way slowly changing dimensions are handles, data could be loaded in different ways
 - Type 1: update existing records
 - Type 2: add new record and update existing records to expire it
 - Type 3: update existing record with new value and old value
 - Type 4: update main dimension mini-dimension appropriately
- Most commercial mature ETL tools handle data loading to SCD tables
 - We just have to configure the ETL flow as we want!

Handling Surrogate Key

- Primary key of dimension tables are called which is a separate integer based auto incrementing column
 - Separated from business/natural keys being received from source systems
 - the keys in the transaction source systems
- When loading data to dimension tables, surrogate key must be,
 - auto populated by the data warehouse, or
 - inserted by the ETL flow
- When loading data to fact tables correct surrogate key in referencing dimension tables must be looked up and inserted in to the fact tables' reference columns

Loading into Fact Tables

- Fact tables are usually larger in size compared to dimension tables and thus, only changes (modifications of the source systems) are loaded
- Based on the type of the fact table, different loading approaches will be used
 - Transaction: mostly insert new transactional records
 - Periodic snapshot: mostly insert new aggregated records based on some date period
 - Daily snapshot, weekly snapshot
 - Accumulating fact tables: insert new events and update existing events
- Based on the behaviour of source systems, different loading approaches may have to be used
 - If source system updates old transactional records, a similar approach may have to be followed when loading transactional data to fact tables
- Foreign keys must be properly dealt with using surrogate keys
 - Correct surrogate key in relevant dimension table must be referenced (looked up) and inserted to the fact table

Loading into Aggregations

- Aggregate tables improve the performance of the data warehouse (specially when the data volume is high) when data is being consumed by BI layer components such as reports, dashboards
 - Rather than performing required aggregation calculation dynamically to generate the report/dashboard, these aggregate (summarized) values are pre-calculated and stored in the data warehouse
- ETL tool populates and maintains aggregate fact tables
 - Once the base fact tables are loaded, aggregate tables can be populated
- Fastest way is to incrementally load the aggregate tables
 - May require a delete-insert approach for some records
- Aggregates must always be consistent with base data (main fact table)

Process Flow Automation

Ochestration and Automation Tools

- Every enterprise DW should have a robust ETL orchestration and automation tool(s) that manage the entire ETL process through a metadata-driven job control environment
- It should be aware of and control the relationships and dependencies between ETL jobs/tasks
- Should support a fully automated process
 - This includes managing ETL processes during failures
- In general, orchestration and automation tools should support following features:
 - Task orchestration: define relationships and dependencies between ETL jobs/tasks
 - Triggering of ETL jobs: executing ETL jobs based on a set date-time or based on some other trigger such as a file
 - Metadata capture
 - Logging
 - Notifications